Single-Site Rigidity and a Sporadic Exception for Prime-Indexed Nörlund–Stirling Quotient Polynomials

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Abstract

For $p \equiv 1 \pmod 8$ let $f_p = q_{(p-5)/4}$ and $g_p = q_{(p-3)/2}$ be the primitive odd-index Nörlund–Stirling quotient polynomials. We investigate whether a simultaneous zero at one prescribed negative integer $-a$ forces a whole prefix of simultaneous zeros at $-2, -3, \dots, -a$. We establish an effective large-prime rigidity theorem when a finite family of explicitly defined weights has no zero-sum proper subset. An exact meet-in-the-middle certificate verifies this hypothesis for every $2 \le a \le 28$. We then completely classify *every* eligible prime for $2 \le a \le 12$: the prefix implication holds with precisely one exceptional pair, $(a, p) = (10, 3617329)$, where both polynomials vanish at -10 but do not share the root -2. The exceptional cancellation is exhibited by an exact Gaussian moment equal to $-1560p$. The exception proves that a universal unqualified single-site-to-prefix principle is false. For each fixed $2 \le a \le 28$, the natural density of primes with a simultaneous zero at $-a$ remains exactly $1/(2 \cdot 4^{\pi(a)})$. Classical Stirling identities and quartic Kummer theory are the inputs. We claim the single-site forcing theorem, its effective exceptional-prime bound, and the complete small-site classification with a genuine non-prefix exception as the specific new contributions of this paper.

Keywords: Nörlund–Stirling quotients; simultaneous polynomial zeros; prime-modulus congruences; quartic residues; exceptional primes; Chebotarev density.
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1 Introduction and normalization

Put

$$S_n(X) = [z^n]\Big(\frac{z}{1-e^{-z}}\Big)^X, \qquad S_n(X) = \frac{X^2(X-1)}{D_n} q_n(X) \quad (n \ge 5 \text{ odd}),$$

where $q_n \in \mathbb{Z}[X]$ is primitive and D_n is the corresponding positive integer normalizer. The good-prime normalization needed below is proved directly: if $p > n+1$, then $p \nmid D_n$, as explained after the definition of the prime-indexed pair. For clarity our prime-indexed pair is

$$p \equiv 1 \pmod 8, \quad p \ge 41, \quad h = (p-1)/4, \quad f_p = q_{h-1} = q_{(p-5)/4}, \quad g_p = q_{2h-1} = q_{(p-3)/2}.$$

The indices are odd and satisfy the good-prime inequality $p > 2(2h-1) = p-3$. For a fixed negative centre $-a$ we impose $2 \le a < (p+3)/2$; this implies $(2h-1+a)!$ is a p-unit, and $a < p$, so the negative-centre Stirling identity transfers to the primitive quotients without hidden divisions by p.

To justify the normalization, write $H(z) = (1-e^{-z})/z = \sum_{k\ge 0}(-1)^k z^k/(k+1)!$. If $p > n+1$, then H, its inverse $A = 1/H$, $\log A$ and $\exp(X \log A)$ are coefficientwise in $\mathbb{Z}_{(p)}[X]$ through degree n: every denominator introduced by the truncated logarithmic and exponential series is a product of integers smaller than p. Thus $S_n(X) \in \mathbb{Z}_{(p)}[X]$. For odd $n \ge 5$, the identities

$\log A(z) = z/2 +$ even powers of z and $A(z) = 1 + z/2 +$ even powers of z show that S_n has a double zero at $X = 0$ and a zero at $X = 1$. Dividing by the monic $X^2(X-1)$ preserves p-integrality. Since q_n is primitive over $\mathbb{Z}$ and $S_n/[X^2(X-1)] = q_n/D_n$, Gauss's lemma implies $p \nmid D_n$.

Stirling exponential sums, their congruences, quartic power residues, and Chebotarev's theorem are classical; see [1, 2, 3, 4, 8]. The assertion studied here is more specific: *does a common root at one prescribed site force the entire preceding consecutive block?* The answer is asymptotically yes for an explicit finite range of sites, but not literally always: we prove that a unique exceptional pair occurs for $2 \le a \le 12$. The broader arithmetic setting was also considered in a separate CCR-k refinement article by the author [7]. The source-independent primitive quotients used here are defined from scratch and are *not* identified with a source-deformed transport polynomial from that work. No result of [7] is needed in our proofs.

The precise distinction from earlier prescribed-prefix work is logical: previously one *assumed* that all negative centres in a block were common roots; here only the single value at $-a$ is assumed. The question is whether a simultaneous zero at one prescribed site forces a full earlier prefix. The complete conclusion turns out to require a sporadic exception. Our principal contributions are:

1. An integer-weight no-cancellation criterion implying isolated-site-to-prefix forcing for all sufficiently large good primes, with an explicit effective numerical threshold.
2. An exact finite certificate of this criterion at all sites $2 \le a \le 28$.
3. A complete classification for every eligible prime at sites $2 \le a \le 12$, containing exactly one counterexample to unqualified forcing, at $(a, p) = (10, 3617329)$.
4. Exact natural densities of the resulting isolated common-root events, and a sharp local resultant example separating collision multiplicity from p-adic depth.

We regard the finite-difference, Gaussian-integer, and Kummer inputs as classical. The proposed novelty is the *specific simultaneous-root forcing and exceptional-pair classification* for the prime-indexed quotient family. To the best of the author's knowledge, this single-site forcing theorem and the exhaustive sporadic-exception classification do not appear in the cited literature.

2 Finite-difference reduction

For integers $a \ge 2$, put

$$w_{a,b} = (-1)^{a-b}\binom{a}{b}b^{a-1} \quad (1 \le b \le a), \qquad \sum_{b=1}^{a} w_{a,b} = 0.$$

The last equality is the a-th finite difference of a polynomial of degree $a-1$. Write $\{{N \atop a}\}$ for the Stirling number of the second kind.

Lemma 2.1 (Exact negative-centre identity)**.** *For every odd $n \ge 5$,*

$$-(n+a)!S_n(-a) = a!\left\{{n+a \atop a}\right\} = \sum_{b=1}^{a}(-1)^{a-b}\binom{a}{b}b^{n+a}.$$

Consequently, for $n_j = jh - 1$ and $j = 1, 2$,

$$-(n_j+a)!S_{n_j}(-a) = \sum_{b=1}^{a} w_{a,b}(b^h)^j.$$

Proof. Write $S_n(-a) = [z^n]((1-e^{-z})/z)^a$ and expand the numerator binomially; for odd n the sign is negative. The second equality is the standard Stirling exponential sum [1, §26.8]. Substitute $n_j + a = jh + a - 1$ for the final identity. □

Let $\mu_4 = \{1, -1, i, -i\} \subset \mathbb{Z}[i]$ and suppose $p \equiv 1 \pmod 8$. For each prime $r \le a$, choose $e_r \in \{0, 1, 2, 3\}$ such that $r^h = i_p^{e_r}$ for a fixed primitive fourth root $i_p \in \mathbb{F}_p$. Extend the digits additively to every $b \le a$ by prime factorization, giving $b^h = i_p^{e_b}$. Form the two Gaussian moments

$$E_j(e) = \sum_{b=1}^{a} w_{a,b} i^{j e_b} \in \mathbb{Z}[i], \qquad j = 1, 2.$$

If $f_p(-a) = g_p(-a) = 0 \pmod p$, then $E_1(e), E_2(e)$ vanish under $\mathbb{Z}[i] \to \mathbb{F}_p$ sending $i \mapsto i_p$, and hence p divides their respective Gaussian norms. This implication is elementary, but the use of *two* moments is essential.

Lemma 2.2 (No-cancellation criterion)**.** *Suppose no nonempty proper subset $I \subset \{1, \dots, a\}$ has $\sum_{b \in I} w_{a,b} = 0$ in $\mathbb{Z}$. Then the only assignment of fourth-root digits for prime bases $r \le a$ for which $E_1(e) = E_2(e) = 0$ as Gaussian integers is the all-zero assignment.*

Proof. Let A, B, C, D be the sums of weights in the four classes where i^{e_b} respectively equals $1, -1, i, -i$. Then $E_1 = (A - B) + i(C - D)$ and $E_2 = A + B - C - D$. With $A + B + C + D = 0$, the equations $E_1 = E_2 = 0$ imply $A = B = C = D = 0$. Every nonempty class is therefore a subset of zero weight. The hypothesis permits only one nonempty class; because $b = 1$ occurs with $1^h = 1$, this is the class A, and all $b^h = 1$, hence all prime-base digits vanish. □

3 An effective single-site rigidity theorem

Define the integer

$$T_a = \sum_{b=1}^{a} |w_{a,b}| = \sum_{b=1}^{a} \binom{a}{b} b^{a-1}.$$

Theorem 3.1 (Effective single-site rigidity)**.** *For any fixed $a \ge 2$ satisfying the no-cancellation hypothesis of Lemma 2.2, every prime $p \equiv 1 \pmod 8$ with*

$$p > \max\{41,\, 2a + 1,\, T_a^2\}$$

satisfies the equivalences

$$\begin{aligned} f_p(-a) \equiv g_p(-a) \equiv 0 \pmod p &\iff r^{(p-1)/4} \equiv 1 \pmod p \quad \text{for every prime } r \le a \\ &\iff f_p(-b) \equiv g_p(-b) \equiv 0 \pmod p \quad \text{for all } 2 \le b \le a. \end{aligned}$$

In particular the exceptional primes are not merely finite: each is bounded above by the explicit number $\max\{41, 2a + 1, T_a^2\}$ (outside the stated normalization eligibility).

Proof. Suppose the two values at $-a$ vanish, and encode the actual quartic residues as above. If the digit assignment is nonzero, Lemma 2.2 ensures at least one E_j is nonzero as a Gaussian integer. Its norm is a positive integer at most T_a^2, because $|E_j| \le T_a$. But p divides that norm, contradicting $p > T_a^2$. Thus all prime bases $r \le a$ are fourth powers mod p.

Conversely, if all the relevant prime bases are fourth powers, then $b^h = 1$ for every $1 \le b \le a$; Lemma 2.1 and $\sum_b w_{a,b} = 0$ show both values vanish at $-a$. The same argument applies to each earlier $-b$. Finally, simultaneous vanishing on the entire prefix implies fourth-power residuosity of all $b \le a$, by applying the first-polynomial identities successively for $b = 2, 3, \dots, a$: the resulting lower triangular matrix has diagonal entries b^{b-1}, units modulo p. □

Proposition 3.2 (Certified no-cancellation range)**.** *The hypothesis of Lemma 2.2 holds for every integer $2 \le a \le 28$.*

Exact finite certificate. For each a split $\{1, \dots, a\}$ into sets of sizes $\lfloor a/2 \rfloor$ and $\lceil a/2 \rceil$. Enumerate the integer subset sums of the left and right halves. An exact match $u + v = 0$, other than the empty subset and the full subset, would violate the assertion. An exact finite enumeration of these two labelled subset-sum families for all 27 values of a finds no prohibited match. The enumeration protocol, including checks against the two trivial subsets, is specified in Section 4. This is a finite computational proof, not an assumed infinite family statement. □

4 Computational proof certificates

The finite calculations are small enough to be specified exactly and rerun without a computer algebra system. They are part of the proof of Proposition 3.2 and of the complete exceptional-prime theorem below, not evidence for an unbounded extrapolation.

Certificate I: integer subset-sum exclusion. For fixed a let $w_b = w_{a,b}$. Divide the indices into two halves L and R and compute the two sets of labelled subset sums

$$\mathcal{S}_L = \{(\sum_{b\in I} w_b, I) : I \subseteq L\}, \qquad \mathcal{S}_R = \{(\sum_{b\in J} w_b, J) : J \subseteq R\}.$$

Match $u \in \mathcal{S}_L$ against $-u \in \mathcal{S}_R$. Accept only the two trivial zero-sum subsets, the empty set and the full index set. All operations are on unbounded Python integers. The enumeration was performed for every $a = 2, \ldots, 28$. The labels I, J are retained, so coincident sums from different subsets are not accidentally discarded.

Certificate II: exception exhaustiveness. For each $a = 2, \ldots, 12$ enumerate all $4^{\pi(a)} - 1$ nontrivial vectors $e = (e_r)_{r\le a}$ with $e_r \in \{0, 1, 2, 3\}$. Multiplicativity supplies e_b for each $b \le a$. Compute the pairs $(E_1(e), E_2(e)) \in \mathbb{Z}[i]^2$, then

$$G(a, e) = \gcd\big(N(E_1(e)), N(E_2(e))\big), \qquad N(x + iy) = x^2 + y^2.$$

Factor every positive $G(a, e)$ into prime factors and certify the factorization by exact division with a primality test. Retain only $p \equiv 1 \pmod 8$ satisfying the size and normalization conditions; recompute actual residues $r^{(p-1)/4} \pmod p$ and both original Stirling sums for each surviving pair (a, p). This final step is crucial: dividing a Gaussian norm is only a *necessary* test, since it does not prescribe the correct choice of fourth root modulo p. The same exact-integer procedure was run for every site $2 \le a \le 12$, including separate implementations of the Gaussian moment calculation and the final modular test.

Independent arithmetic audit. A second implementation evaluates the second-kind Stirling numbers by the exact recurrence

$$\left\{{N \atop k}\right\} = k \left\{{N-1 \atop k}\right\} + \left\{{N-1 \atop k-1}\right\}$$

modulo the selected prime. It agrees with the finite exponential sums on the independently checked sample cases. The candidate-prime factor lists are given in Table 1; the finite enumeration is completely determined by the mathematical procedure above.

Validation scope and checks. For each a, the number of nontrivial fourth-root assignments is exactly $4^{\pi(a)} - 1$, as reported in Table 1. The finite exceptional-prime search is therefore *complete by construction*: every potential failure must divide at least one positive Gaussian norm gcd, and every such gcd is factored completely and reconstructed from its prime factors before the actual modular moments are checked. The tests use exact unbounded integers; no numerical tolerance, random prime sample, or preselected prime cutoff enters the exceptional-prime classification. A separately implemented Gray-code subset-sum search and independent Gaussian-moment factorization reproduced the $a \le 28$ no-cancellation and $a \le 12$ exception outcomes. The elementary finite-field and Stirling calculations were also checked independently at selected values. The exact arithmetic methodology and its verification scope are stated here.

5 Exact exceptional-prime classification through twelve sites

Theorem 5.1 (Complete small-centre classification)**.** *For every integer* $2 \le a \le 12$ *and every prime* $p \equiv 1 \pmod 8$ *with* $p \ge 41$*, the following assertions are equivalent,* except *for the unique parameter pair* $(a, p) = (10, 3617329)$*:*

1. $f_p(-a) \equiv g_p(-a) \equiv 0 \pmod p$*;*
2. $r^{(p-1)/4} \equiv 1 \pmod p$ *for every rational prime* $r \le a$*;*
3. $f_p(-b) \equiv g_p(-b) \equiv 0 \pmod p$ *for every* $2 \le b \le a$*.*

At $(a, p) = (10, 3617329)$*, assertion (1) holds whereas (2) and (3) fail. In particular, this exception is genuine and cannot be removed from the theorem.*

Exhaustive finite certificate. The implication (2)⇔(3) follows from the lower-triangular Stirling system, and (2)⇒(1) is immediate from the finite-difference formula. It remains to classify violations of (1)⇒(2). For a fixed a enumerate the $4^{\pi(a)} - 1$ nonzero fourth-root digit assignments on primes $r \le a$. For each, compute the exact Gaussian moment pair (E_1, E_2) and

$$G(a, e) = \gcd\big(N(E_1(e)), N(E_2(e))\big) > 0.$$

The positivity follows from Proposition 3.2 and Lemma 2.2. A counterexample prime necessarily divides one of these finitely many positive integers. The exact finite computation lists all distinct prime factors, tests their primality, and verifies that repeated division by the listed factors leaves quotient one *for every assignment*. It then checks each eligible $p \ge 41$, $p \equiv 1 \pmod 8$ by modular exponentiation and the two explicit Stirling sums. For $a = 2, \dots, 8$ the previous verified certificate returns no exceptions. The new certificate for $a = 9, \dots, 12$ yields Table 1; the sole true counterexample is $(10, 3617329)$. No untested prime can be an exception because the norm-gcd divisors are exhaustively certified. □

a	Eligible Gaussian-norm candidate primes	Genuine exceptions	Assignments
2–4	none	none	$3, 15, 15$
5	41	none	63
6	$73, 257$	none	63
7	$233, 1873, 2953$	none	255
8	$73, 7457$	none	255
9	$52249, 132929$	none	255
10	$241, 313, 6569, 32993,$ $489001, 1097513, 3617329, 9803873$	3617329	255
11	$281, 337, 433, 2161, 8353, 40193,$ $83497, 375049, 400601, 504457,$ $12119761, 42074561$	none	1023
12	$89, 137, 353, 617, 641, 1033,$ $1409, 2273, 409217, 1424809,$ $2186113, 3448561, 59951393, 182991329$	none	1023

Table 1: Complete eligible exception-candidate lists, not a bounded brute-force search over primes. For each fixed a the assignment count is exactly $4^{\pi(a)} - 1$; certification is by exact norm divisibility.

Proposition 5.2 (An explicit non-prefix simultaneous zero)**.** *Let* $p = 3617329$ *(prime),* $h = (p-1)/4 = 904332$*, and*

$$f_p = q_{904331}, \qquad g_p = q_{1808663}.$$

Then $f_p(-10) \equiv g_p(-10) \equiv 0 \pmod p$*, but* $f_p(-2) \not\equiv 0 \pmod p$*.*

Proof. Direct modular exponentiation gives

$$2^h \equiv -1, \quad 3^h \equiv 1, \quad 5^h \equiv -1, \quad 7^h \equiv -1 \pmod{p}.$$

Consequently $b^h \equiv \pm 1 \pmod{p}$ for every $1 \le b \le 10$. Applying the exact weights $w_{10,b}$ and replacing these residues by their integer signs, the two finite Gaussian-moment representatives are

$$E_1 = \sum_{b=1}^{10} w_{10,b} \operatorname{sgn}(b^h) = -5643033240 = -1560p, \qquad E_2 = \sum_{b=1}^{10} w_{10,b} = 0.$$

Here $\operatorname{sgn}(b^h)$ means the representative $+1$ or -1 of $b^h \pmod{p}$; E_j is a *finite representative modulo* p, not the original large Stirling integer. The reduction in Lemma 2.1 makes both quotient values zero. But at -2 that lemma gives $-(n_1+2)!S_{n_1}(-2) = 2(2^h-1) \equiv -4 \not\equiv 0 \pmod{p}$. All normalizations and factorials are p-units, so $f_p(-2)$ is nonzero as asserted. Primality of p is independently certified by deterministic trial division (up to $\lfloor\sqrt{p}\rfloor$). □

6 Exact natural densities and first failure

Let $\pi(a)$ count rational primes $\le a$, and let

$$K_a = \mathbb{Q}(\zeta_8,\ r^{1/4} : r \le a \text{ rational prime}).$$

Lemma 6.1 (Quartic Kummer degree)**.** *For every $a \ge 2$, $[K_a : \mathbb{Q}] = 2 \cdot 4^{\pi(a)}$.*

Proof. The cyclotomic field $F = \mathbb{Q}(\zeta_8) = \mathbb{Q}(i, \sqrt{2})$ has degree 4. Adjoining $2^{1/4}$ has relative degree 2: $\sqrt{2} \in F$, whereas $2^{1/4} \notin F$, since the nonnormal quartic field $\mathbb{Q}(2^{1/4})$ cannot lie in the abelian Galois field $F/\mathbb{Q}$. The resulting field ramifies only at 2. On successively adjoining $r^{1/4}$ for distinct odd primes r, the intermediate field is unramified at r: its generators have discriminants supported on 2 and previously adjoined primes. At a prime above r, $v(r) = 1$, so $X^4 - r$ is Eisenstein. Each odd radical therefore raises the degree by 4, giving $4 \cdot 2 \cdot 4^{\pi(a)-1}$. □

Corollary 6.2 (Density of a single prescribed common root)**.** *For each fixed $2 \le a \le 28$, the set of primes $p \equiv 1 \pmod{8}$ for which $f_p(-a) \equiv g_p(-a) \equiv 0 \pmod{p}$ has natural density*

$$\boxed{\delta_a = \frac{1}{2 \cdot 4^{\pi(a)}}.}$$

Moreover, the first missing root in the initial consecutive block $-2, -3, \ldots$ can occur only at a rational-prime centre, apart from finitely many ineligible small primes; for odd prime r the exact absolute density of primes whose first missing centre is $-r$ equals $3/(8 \cdot 4^{\pi(r-1)})$.

Proof. By Theorem 3.1 and Proposition 3.2, except for finitely many primes the single-site common-root property is equivalent to every rational prime $r \le a$ being a fourth power modulo p. This holds exactly when p splits completely in K_a, which has Chebotarev density $1/[K_a : \mathbb{Q}]$. For the prefix the same equivalence holds for every fixed endpoint, even without the no-cancellation hypothesis, by the triangular Stirling system. A composite m cannot be the first failure: all prime factors of m are smaller and would already be fourth powers. The density at an odd prime r is $\delta_{r-1} - \delta_r = \frac{3}{4}\delta_{r-1}$; here δ_m denotes prefix density, defined at all m, not just $m \le 28$. □

Remark 6.3 (Scope of the result)**.** *The single-site theorem applies to each fixed $a \le 28$ outside an explicit finite prime set, and is exception-free for $a \le 12$ except for the one explicitly certified pair $(a, p) = (10, 3617329)$. It makes no statement about a growing with p. The prime-jump distribution is related to the classical least quartic nonresidue problem; the proposed specialized contribution here is the implication from a* single prescribed common-root site *to an entire root prefix, together with the exact sporadic failure, not the classical quartic-residue distribution itself. The methods used are standard Gaussian linear algebra and Kummer–Chebotarev theory. No novelty is claimed for the underlying classical power-residue distribution or Kummer-theoretic methods.*

7 A sharp local resultant example and scope

For $p = 577$ the two indices are 143 and 287, and both polynomials have the three common roots $-2, -3, -4$ modulo 577. A complete modular coefficient computation gives

$$\gcd_{\mathbb{F}_{577}[X]}(\bar{q}_{143}, \bar{q}_{287}) = (X+2)(X+3)(X+4).$$

Each of these three roots is simple in both reductions. Direct first-order Hensel calculations show that the lifts for q_{143} and q_{287} differ at every one of the three centres; the six respective first digits are $(514, 219, 156)$ and $(412, 319, 171)$. Hence each of the three common residual factors contributes exactly one to the local resultant valuation; all other factors contribute zero, because the displayed gcd is the entire common residual factor. Since the leading coefficients are 577-adic units, this proves

$$v_{577}(\mathrm{Res}(q_{143}, q_{287})) = 3.$$

These exact modular gcd and first-Hensel-digit calculations were independently recomputed, using integer or finite-field arithmetic. The values stated above specify their outputs. This example distinguishes three depth-one collisions from a single root of depth three.

The finite verification has three parts: exact subset-sum enumeration for $a \le 28$, complete Gaussian-norm prime-candidate enumeration and modular filtering for $a \le 12$, and independent Stirling-recurrence checks at sample parameters. These calculations are described in Section 4. Independent exact-arithmetic implementations reproduce the first two enumerations and the $p = 577$ Hensel calculation. No finite computation is claimed to extrapolate beyond the stated range.

8 Open Problems and Future Directions

Unsolved problem. For distinct odd $5 \le r < s$, does there exist a good prime $p > 2s$ and an $X \in \mathbb{Z}/p^4\mathbb{Z}$ such that $q_r(X) = q_s(X) = 0 \pmod{p^4}$? The present mod-$p$ work does not settle this question. A separate nonvanishing hypothesis for first Hensel derivatives is also deliberately left as a conjecture, not used here.

9 Related literature, originality and mathematical contribution

The identity of Lemma 2.1, congruence periodicity of Stirling sequences, Gaussian integers, Kummer extensions, and Chebotarev are standard tools. Earlier work by Adelberg [2, 3] studies finite differences, factorization and arithmetic of Nörlund polynomials, while Young [4] develops Kummer-type congruences involving Stirling numbers. Kwong [5] studies periodicities modulo integers, and Sun [6] establishes broad combinatorial congruences for Stirling numbers. None of these classical ingredients alone is claimed as new.

The paper's proposed contribution is the *single-site-to-prefix forcing theorem* for a specific prime-dependent pair of primitive quotient polynomials, its explicit finite-prime effective threshold, and the exhaustive small-site classification exhibiting one genuine sporadic, non-prefix collision. The latter shows why an unqualified all-prime rigidity conjecture is false. The density consequence is retained for completeness, although its power-residue distribution mechanism is classical.

Contribution and novelty statement. To the best of the author's knowledge, the precise *single-site-to-prefix* implication for the good-prime pair $q_{(p-5)/4}, q_{(p-3)/2}$, its explicit Gaussian-norm exceptional-prime bound, and the complete classification through $a = 12$ with the non-prefix exception $(10, 3617329)$ have not been previously recorded together in the cited Nörlund–Stirling literature. These constitute the specific originality claim of this paper. The underlying Stirling exponential sum, fourth-power-residue conditions, Gaussian norm, Kummer extension, and Chebotarev theorem are *not* claimed as new. The novelty claim is confined to the explicitly

formulated quotient-polynomial rigidity and exception theorems, rather than to their classical algebraic ingredients.

Statements and Declarations

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Ethics approval and consent to participate. Not applicable.
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